

Go Green Cooper!

Spring 2026

VIP Smart Cities & ME493 Climate Action Planning

Ray Ramadhar, Ayushi Bhattacharjee , Danielle Chiu, Jhanjaneth Perea, Hyungkyu (Tom) Lee

Prof. Baglione & Prof. Rundell

Cooper Campus Wide Emissions

tCO_2e based on 2025 updated annual NYC CC coefficients

2024 Total Emissions

3,541 tCO_2e

Contribution by Source

66.5% Electricity

33.5% Natural Gas

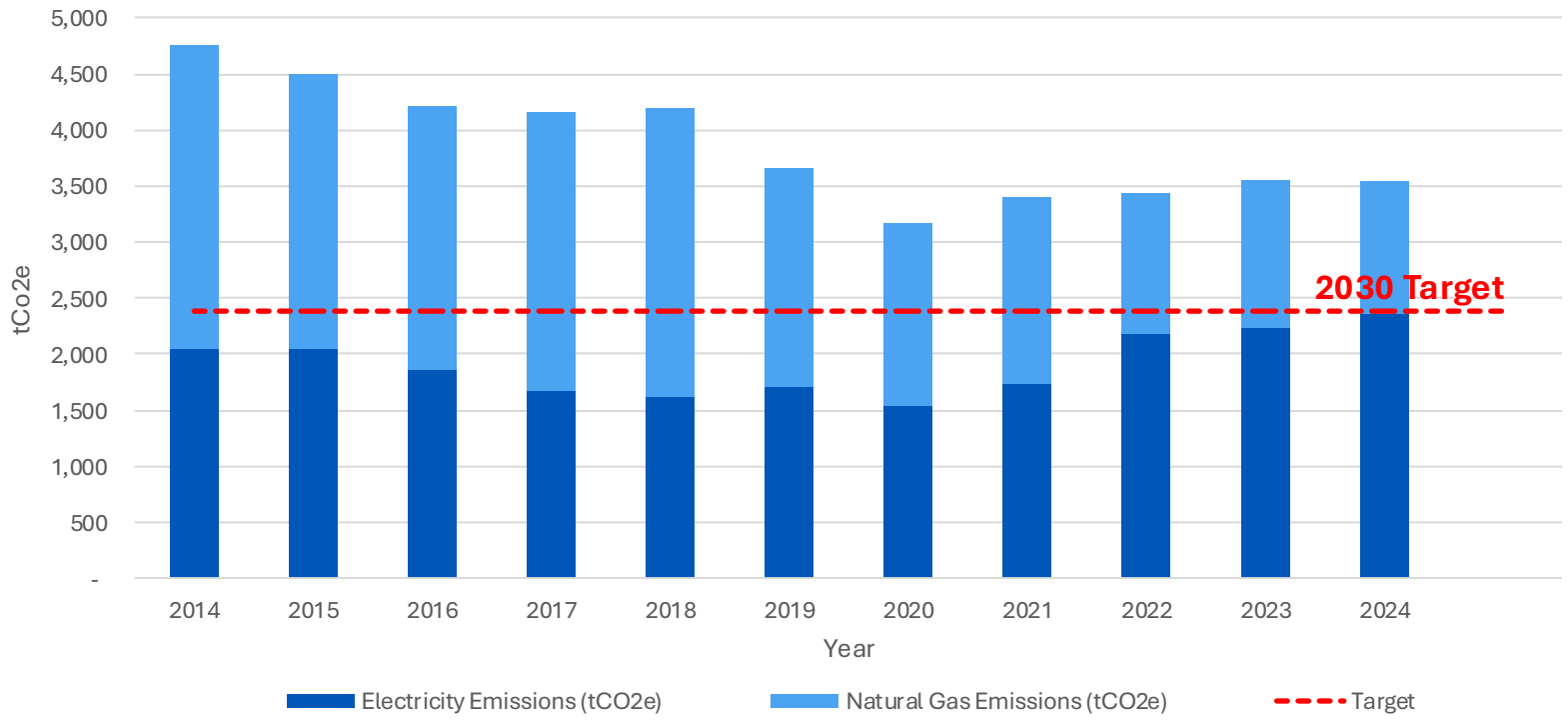
Contribution by Building

58% 41Cooper Square

32% Foundation Building

10% Residence Hall

Campus Total Emissions by Source Trends (tCO_2e)



Our Commitment

50 % reduction in Emissions by 2030
- NYC Carbon Challenge

Goal: Reach

2,367 tCO_2e

by 2030

NYC Climate Mobilization Act

Local Law 97 Requires buildings in NYC over 25,000 sqft to reach zero net carbon emissions by 2050

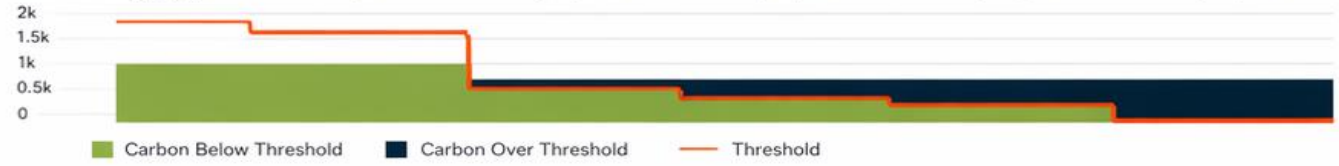
All Cooper Buildings are predicted to face fines in the 2030s due to Local Law 97 penalties.

1 **Foundation Building**

Old School control and efficiency measures

Annual Carbon Threshold Summary

tCO_2e/yr	2024–2029	2030–2034	2035–2039	2040–2049	2050+
Emissions	989	655	655	655	655
Threshold	1,877	399	235	34	0
Est. Penalty ($/yr)	$0	$68,610	$112,608	$166,395	$175,610



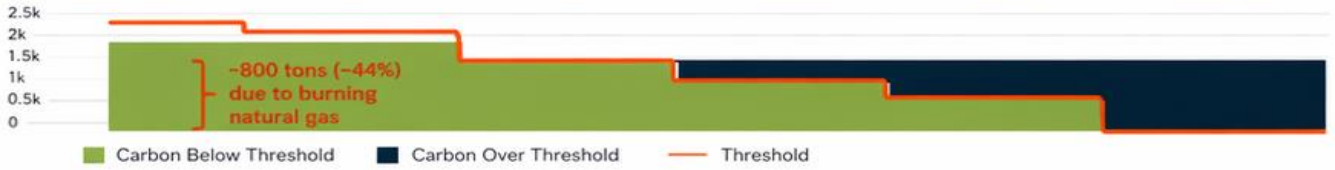
$68,610 / yr
LL97 penalties expected by 2030

2 **41 Cooper Square**

High Natural Gas Electricity emissions

Annual Carbon Threshold Summary

tCO_2e/yr	2024–2029	2030–2034	2035–2039	2040–2049	2050+
Emissions	1,839	1,322	1,322	1,322	1,322
Threshold	2,335	1,387	995	577	0
Est. Penalty ($/yr)	$0	$0	$87,661	$199,674	$354,310

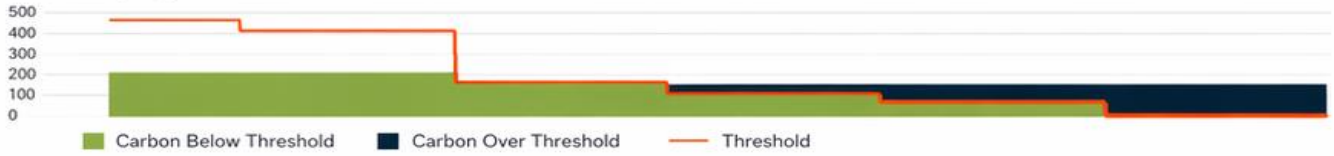


$87,661 / yr
LL97 penalties expected by 2035

3 **Residence Hall**

Annual Carbon Threshold Summary

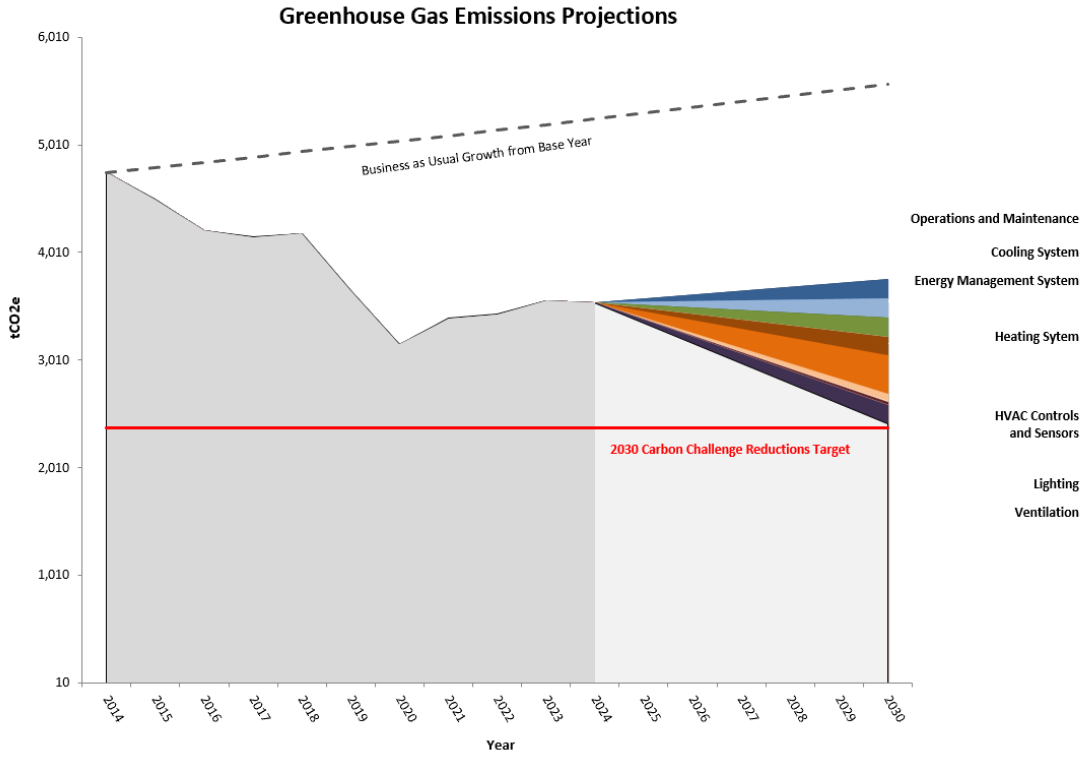
tCO_2e/yr	2024–2029	2030–2034	2035–2039	2040–2049	2050+
Emissions	247	165	165	165	165
Threshold	465	161	96	46	0
Est. Penalty ($/yr)	$0	$885	$18,531	$31,833	$44,138



$885 / yr
LL97 penalties expected by 2035

HVAC optimization is needed to achieve the 2030 Carbon Challenge Reductions target and avoid Local Law 97 penalties.

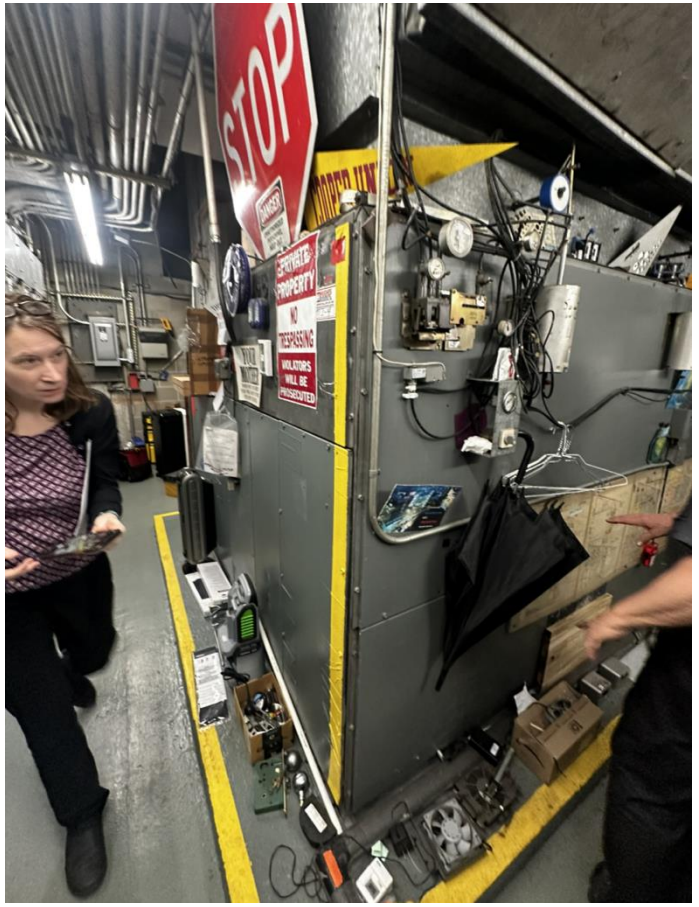
- Retrofitting Opportunities
  - HVAC Optimization -- In Progress
  - Capital Upgrades – Under Evaluation
  - LED Replacement -- Done
- Large drop from 2014-2024
- Slight increase/plateau after 2020

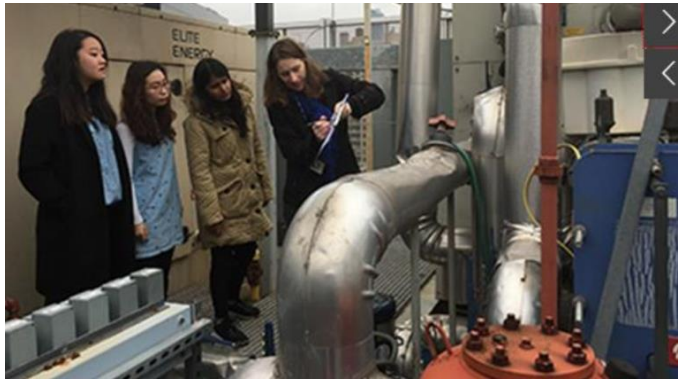


Cooper can reach their energy goals while involving of students and preparing them for the energy field.

Since 2014:

- ~$3M in total energy savings
- > 6800 ton of CO_2 reduction
- 14 Master's thesis projects
- 14 Publications
- Many VIP, Senior Capstone, and IS Projects







41CS Lab Ventilation Audit



Inactive Open Fume Hood in Lab 601



Testing Ventilation in Lab LL219



Point Exhaust





Snorkel Hood On/Off Control

- Checked every lab in 41CS to check vents, fume hoods, and snorkels that are left on - even when not in use
- Summed energy savings from turning off inactive ventilation equipment in labs
- **Savings:**
  - **13,000 CFM**
  - **200 tons of CO_2 emission**
  - **310,000 kWh of electric**

41CS AHU Optimization

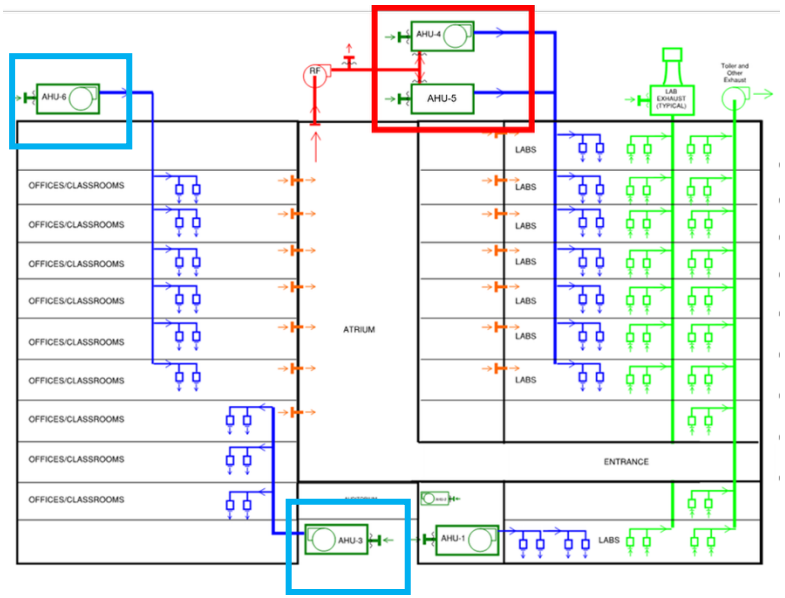


Exterior View of an AHU on 41CS Roof

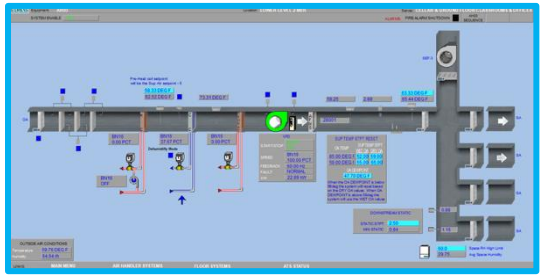
ITEM	LOCATION	AREA SERVED	CFM
AHU - 1	SUB-CELLAR	SUB-CELLAR	22,600
AHU - 2	SUB-CELLAR	CELLAR	4,200
AHU - 3	SUB-CELLAR	CELLAR 1-3 FLOOR	24,600
AHU - 4	ROOF	3-9 FLOOR	29,700
AHU - 5	ROOF	3-9 FLOOR	29,700
AHU - 6	ROOF	3-8 FLOOR	27,500
HV - 1	SUB-CELLAR	MER	3,000

Design Supply Air

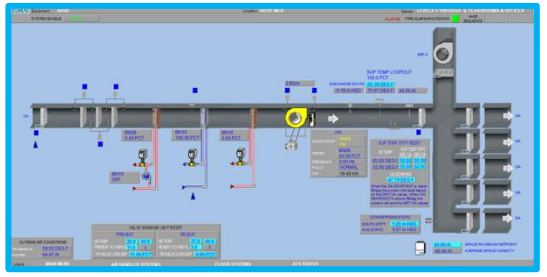
- Air Handler Units (AHUs) in 2026 over supply air
- Created a theoretical model for four AHUs
  - AHU3&6: Service offices/classrooms
  - AHU4&5: Service the 41CS labs
- Baseline model calibrated using trend data
- Improved model calibrated using design values
- Modeled energy and cost savings of **685,000 kWh in electric** and **$200,000/year**



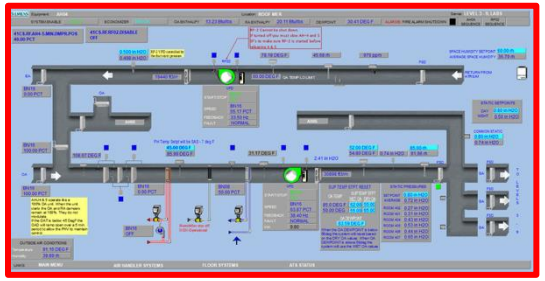
41CS AHU Schematic



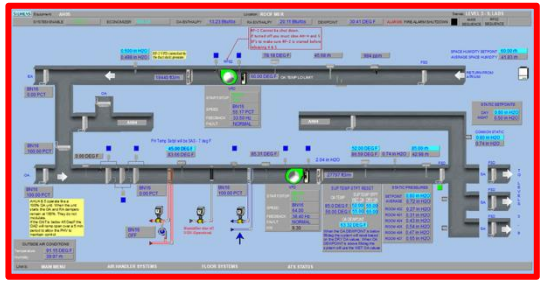
AHU3



AHU6



AHU4



AHU5

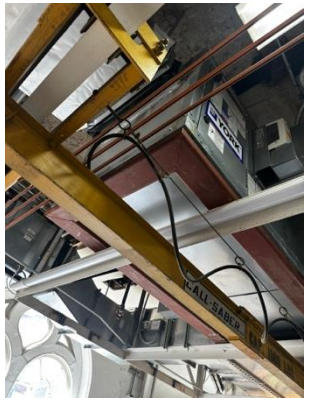
THECOOPERUNION

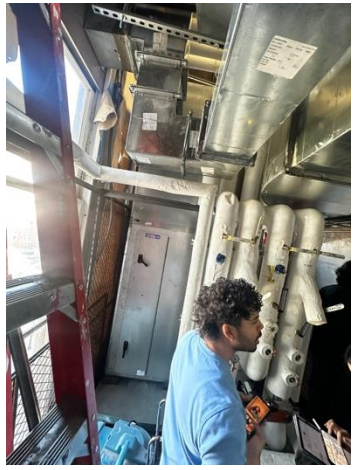
Foundation Building AHU Walkthrough



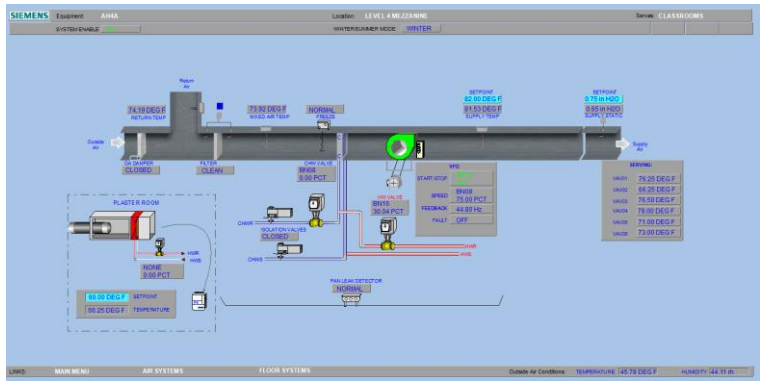








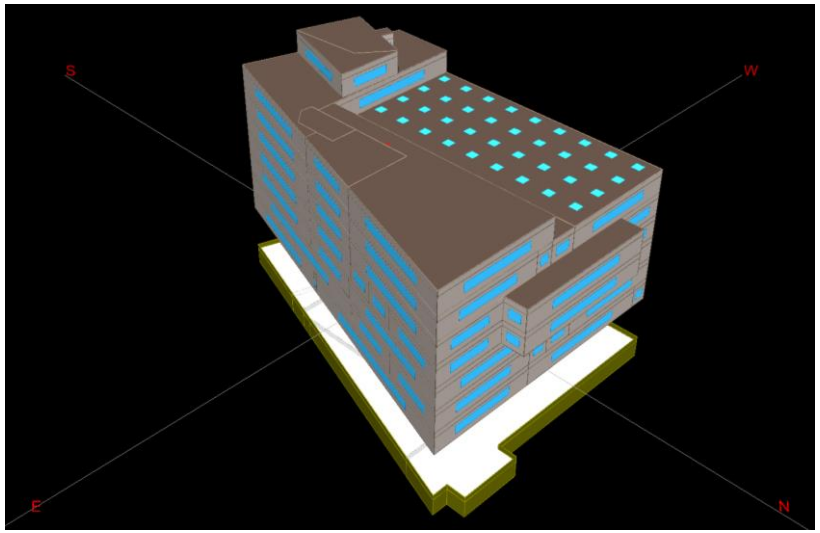
- Traced airflow of AHUs
- Verified if units were mulitzone or single-zone and identified presence of VFDs
- Identified presence of new AHUs or AHUs that underwent replacements not documented
  - AHUs running 24/7
  - Some AHUs have locked speed despite being VFDs



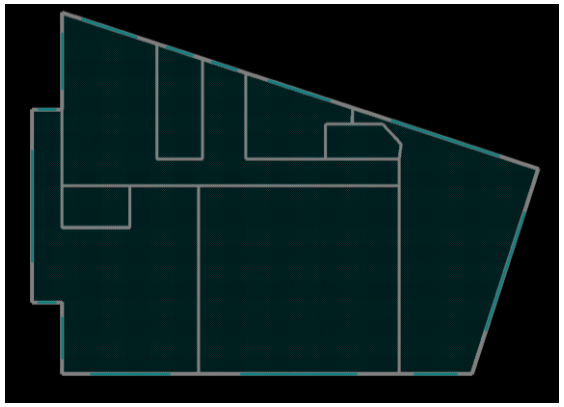
Photos Taken During Foundation Building Walk Through

BMS Screenshots

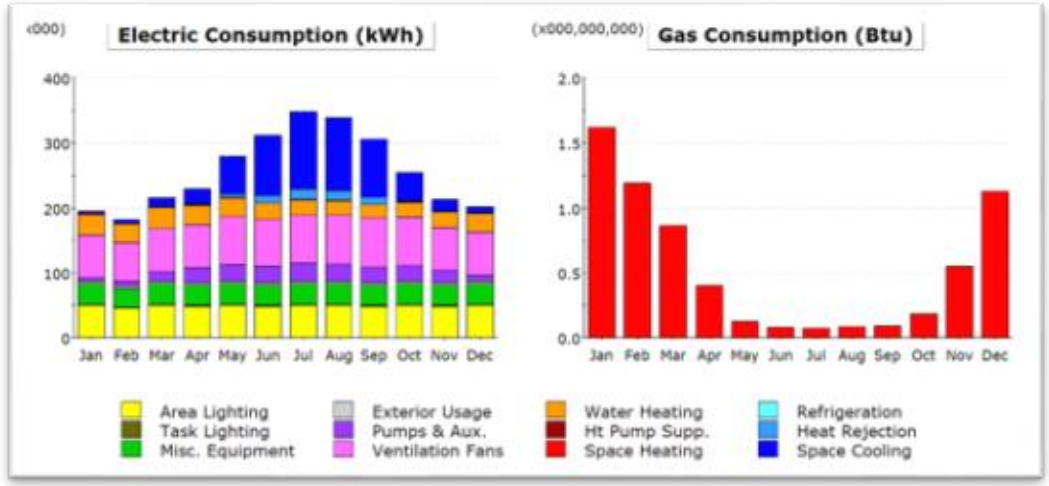
Foundation Building Optimization



E-quest Building Model



E-quest Floor Zoning



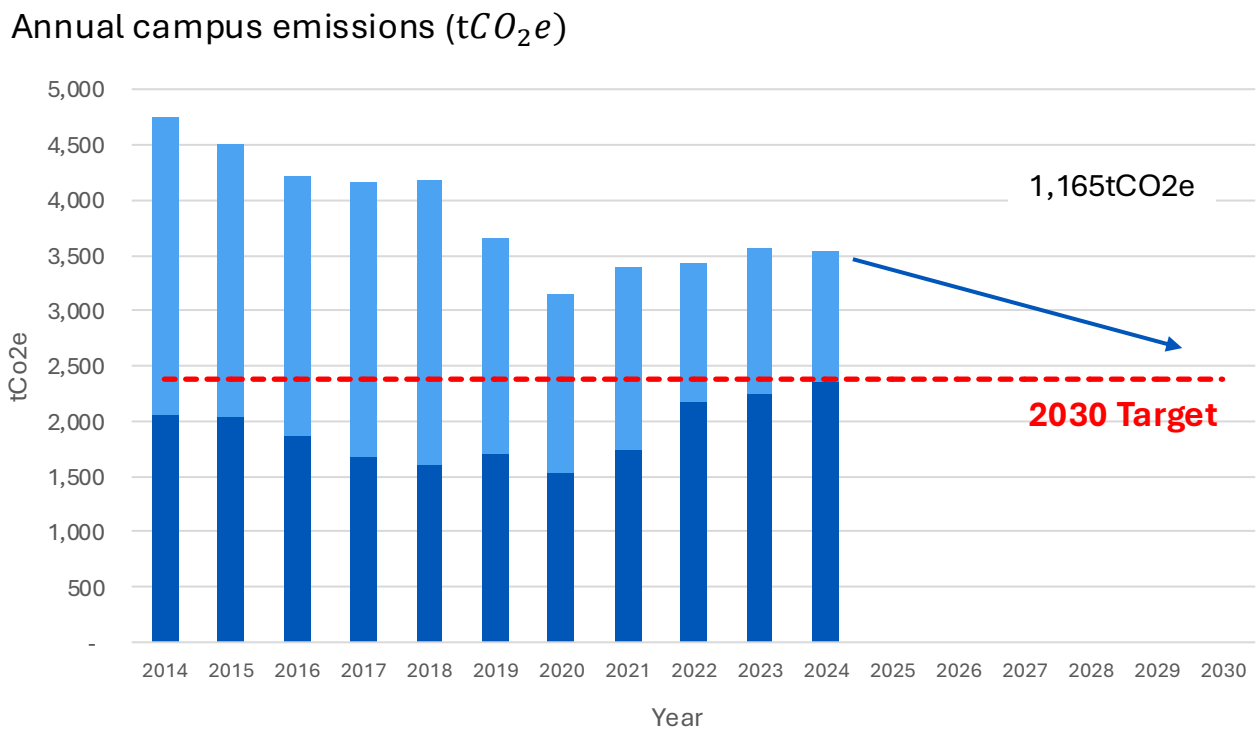
Baseline Simulation Results

- Developed model to align with actual electric and gas consumption values of the Foundation Buildings
- With finalization of baseline model the following ECMs will be implemented to measure savings
  - ECM 1A. Turn off fans from 12 AM -5 AM
  - ECM 1B. 10 PM to 7 AM Space temperature setbacks
  - ECM 2. Operate CV as VAV
  - ECM 3. Convert Multizone Units to VAV

Conclusion: A pathway to a low-carbon campus

Together, we can meet NYC climate goals, achieve Local Law 97 compliance, and build a more resilient campus.

Carbon reduction pathway



How to get involved

Every action counts.
Help Cooper Go Green!



Shut the sash

Keep fume hood sashes closed when not in use.



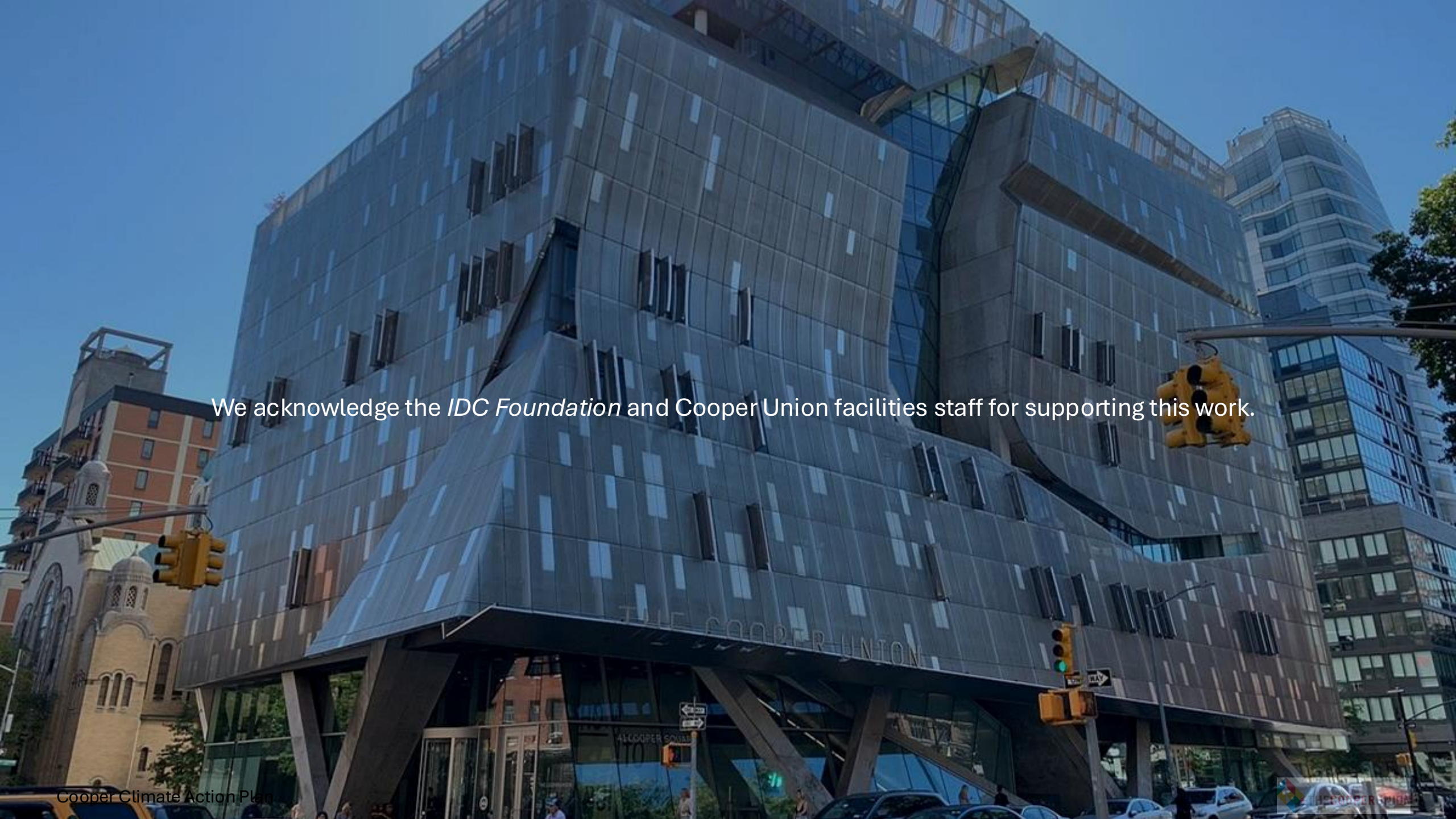
Power down

Turn off lights and equipment when not need.



Get involved in CAP

Join the Cooper Climate Action Plan VIP



We acknowledge the *IDC Foundation* and Cooper Union facilities staff for supporting this work.


Thank You!